

UICT — Uganda
Department of Computer Science

ICT 121

Computer Workshop Practice

Hands-On ICT Skills & Practical Techniques

Ground to Mastery Study Notes

Bravehart
DCS Year 1, Semester 2
Academic Year 2025/2026

Assessment: Coursework 40% | Final Exam 60%

Assessment Breakdown

Component	Weight	Details
Continuous Assessment (Coursework)	40%	Tests, assignments, practicals
Final Examination	60%	Written final exam at semester end

Minimum pass mark: 50% overall

You must pass BOTH components to pass the course unit.

Unit 1: Workshop Safety & Professional Practice

A computer workshop is a controlled environment for assembling, disassembling, diagnosing, and repairing computer hardware. Safety rules protect both the technician and the equipment.

Safety Rules in the Workshop

- Always disconnect power before opening any computer case
- Use an anti-static wrist strap to prevent Electrostatic Discharge (ESD) damage
- Handle components by the edges — never touch circuit traces or chips directly
- Keep liquids away from all electronic equipment
- Use appropriate tools — using the wrong tool can damage screws or components
- Label all cables and components before removing them for easy reassembly
- Work on a clean, well-lit, non-conductive surface

Term	Definition
ESD	Electrostatic Discharge — static electricity that can permanently damage electronic components
Anti-static mat	A grounded mat that dissipates static electricity from the work surface
Anti-static bag	Special packaging used to store and transport sensitive electronic components
PPE	Personal Protective Equipment — safety gear such as goggles and gloves
MSDS	Material Safety Data Sheet — safety information for chemicals used in the workshop

EXAM TIP: ESD can damage components even with voltages too low for humans to feel. A discharge of just 10V can permanently damage a CPU or RAM chip. Always ground yourself before touching internal components.

Unit 2: PC Assembly & Disassembly

PC assembly is the process of building a functional computer from individual components. Following the correct sequence prevents damage and ensures all components are properly installed.

Assembly Sequence

- 1. Prepare workspace and gather tools (Phillips screwdriver, anti-static strap)
- 2. Install CPU into motherboard socket (align triangle marker; do not force)
- 3. Install CPU cooler with thermal paste applied
- 4. Install RAM modules into correct slots (check motherboard manual for dual-channel)
- 5. Mount motherboard into case using standoffs to prevent short circuits
- 6. Install PSU and route cables
- 7. Install storage drives (HDD/SSD) and connect SATA cables
- 8. Install GPU into PCIe x16 slot
- 9. Connect all power cables (24-pin ATX, 8-pin CPU, PCIe)
- 10. Connect front-panel headers (power switch, reset, USB, audio)
- 11. Cable manage for airflow, then close case
- 12. First boot — enter BIOS to verify all components are detected

EXAM TIP: The front-panel header (connecting case buttons to motherboard) is often the trickiest part. The power LED, power switch, reset switch, and HDD LED must all be connected correctly. Consult the motherboard manual for pin layout.

Unit 3: BIOS/UEFI Configuration

BIOS (Basic Input/Output System) or the newer UEFI (Unified Extensible Firmware Interface) is firmware that initialises hardware when the computer starts and provides low-level configuration options.

Setting	Purpose
Boot Order	Sets which device to boot from first (USB, HDD, DVD)
System Time/Date	Sets the real-time clock
XMP/EXPO Profile	Enables RAM to run at its advertised speed rather than default 2133 MHz
Secure Boot	UEFI feature that prevents booting from unsigned/untrusted software
Virtualization	Enables CPU support for running virtual machines (Intel VT-x / AMD-V)
Fan Control	Sets fan curves based on temperature for noise/cooling balance
TPM	Trusted Platform Module — security chip required for Windows 11

Accessing BIOS

- Press Delete, F2, or F10 during POST (varies by manufacturer)
- UEFI supports mouse navigation and has a graphical interface
- BIOS settings are stored in CMOS memory powered by the CMOS battery
- Removing the CMOS battery resets all BIOS settings to factory defaults
- Always save settings (F10 = Save and Exit on most boards) before exiting

EXAM TIP: UEFI replaced BIOS on modern systems. Advantages of UEFI: supports drives larger than 2TB (GPT vs MBR partition scheme), faster boot times, secure boot, graphical interface, and network booting. MBR (BIOS) supports drives up to 2TB only.

Unit 4: Operating System Installation

Installing an operating system is a fundamental workshop skill. A technician must be able to prepare installation media, configure BIOS, partition drives, and complete the OS setup process.

Windows Installation Steps

- 1. Download the OS ISO image from the official source
- 2. Create a bootable USB drive using Rufus or Windows Media Creation Tool
- 3. Set BIOS boot order to boot from USB
- 4. Boot from USB and follow the setup wizard
- 5. Accept license agreement
- 6. Choose 'Custom: Install Windows only' for a clean installation
- 7. Delete existing partitions if doing a fresh install (WARNING: deletes all data)
- 8. Create a new partition on unallocated space; Windows creates system partitions automatically
- 9. Installation copies files and restarts several times
- 10. Configure region, language, user account, and privacy settings
- 11. Install drivers (GPU, network, audio) and run Windows Update

Term	Definition
ISO	A disk image file that contains all files from a CD/DVD/USB installation media
Bootable USB	A USB drive configured to start and run an OS installer
Clean install	Formatting the drive and installing a fresh OS (removes all previous data)
Upgrade install	Installing new OS over existing one, keeping files and settings
MBR	Master Boot Record — older partition scheme; max 2TB, max 4 primary partitions
GPT	GUID Partition Table — modern scheme; supports drives over 2TB, up to 128 partitions

EXAM TIP: Use Rufus with GPT partition scheme and UEFI target system for modern computers. For older systems, use MBR and Legacy BIOS. If Secure Boot is enabled in UEFI, the USB drive must be prepared for UEFI boot.

Unit 5: Driver Installation & Device Management

Device drivers are software programs that allow the operating system to communicate with hardware. After OS installation, installing the correct drivers is critical for all hardware to function properly.

Term	Definition
Driver	Software that allows the OS to interface with a specific hardware device
Device Manager	Windows tool to view, update, disable, and troubleshoot device drivers
Yellow exclamation (!)	Indicates a device driver problem in Device Manager
INF file	Driver information file used during driver installation
Plug and Play (PnP)	Technology that allows hardware to be automatically detected and installed
Driver rollback	Reverting to a previous version of a driver if a new one causes problems
Windows Update	Can automatically download and install many hardware drivers

Driver Installation Priority

- 1. Chipset drivers — must be installed first to enable all other hardware
- 2. Network adapter drivers — enables internet access for downloading remaining drivers
- 3. GPU (graphics card) drivers — from manufacturer website (NVIDIA, AMD, Intel)
- 4. Audio drivers — from motherboard manufacturer website
- 5. Other peripheral drivers — printers, scanners, webcams as needed

EXAM TIP: Always download drivers from the manufacturer's official website. Avoid third-party driver download sites — they often bundle malware. Use Device Manager to check for any devices with missing or problematic drivers (shown with a yellow warning icon).

Unit 6: Network Configuration & Connectivity

Configuring network connectivity is an essential ICT workshop skill. Technicians must be able to set up both wired and wireless connections, assign IP addresses, and troubleshoot connectivity problems.

Setting Up a Wired Network Connection

- 1. Connect Ethernet cable from PC to router/switch
- 2. Ensure NIC driver is installed
- 3. Right-click Network → Open Network & Internet Settings
- 4. For DHCP: set to 'Obtain an IP address automatically'
- 5. For static IP: enter IP address, Subnet mask, Default gateway, DNS server manually

Term	Definition
IP Address	Unique identifier for a device on a network (e.g., 192.168.1.100)
Subnet Mask	Defines which portion of the IP address is the network and which is the host
Default Gateway	The router's IP address; the 'exit point' to other networks/internet
DNS Server	Translates domain names (google.com) to IP addresses
DHCP	Server that automatically assigns IP addresses to devices on the network
ping	Command-line tool to test connectivity to another IP address
ipconfig	Windows command to view IP address, subnet mask, and gateway

EXAM TIP: Troubleshoot network in order: ping 127.0.0.1 (loopback — tests if TCP/IP is working), then ping default gateway (tests local network), then ping 8.8.8.8 (tests internet), then ping google.com (tests DNS). This systematic approach isolates the problem.

Unit 7: Preventive Maintenance & Virus Removal

Preventive maintenance protects computer systems from both hardware and software failures. A key part of software maintenance is protecting systems from malware and viruses.

Term	Definition
Virus	Malicious code that attaches itself to files and replicates when executed
Worm	Self-replicating malware that spreads across networks without user action
Trojan	Malware disguised as legitimate software to trick users into installing it
Ransomware	Malware that encrypts files and demands payment for the decryption key
Spyware	Software that secretly monitors user activities and sends data to attackers
Adware	Software that displays unwanted advertisements; often bundled with free programs
Antivirus	Software that detects, quarantines, and removes malware from a system
Firewall	Security system that monitors and controls network traffic based on rules

Virus Removal Procedure

- 1. Disconnect the infected computer from the network to prevent spread
- 2. Boot into Safe Mode to prevent malware from loading at startup
- 3. Run a full system scan with updated antivirus software
- 4. Quarantine and delete any detected threats
- 5. Use a second-opinion scanner (e.g., Malwarebytes) for thorough cleaning
- 6. Update the OS and all software to patch security vulnerabilities
- 7. Reconnect to network and monitor for recurring symptoms
- 8. If infection persists, consider a clean OS reinstallation

EXAM TIP: Prevention is better than cure. Best practices: keep OS and antivirus updated, use strong unique passwords, do not click suspicious links or email attachments, enable firewall, back up data regularly, and use standard (non-admin) user accounts for daily use.

Unit 8: Printer, Peripheral & Documentation Skills

Peripheral devices extend the functionality of a computer. A workshop technician must be able to install, configure, and troubleshoot printers and other peripherals, and maintain proper documentation of all work done.

Printer Type	Technology	Best For
Inkjet	Sprays liquid ink through tiny nozzles	Photo printing, colour documents, home use
Laser	Uses toner powder fused with heat	High-volume text printing, office environments
Dot Matrix	Pins strike ribbon to form characters	Multi-part forms, receipts, industrial use
Thermal	Uses heat-sensitive paper	Receipts, labels, barcode printing

Documentation Best Practices

- Record every repair: date, computer description, problem reported, diagnosis, action taken, outcome
- Take photos before disassembly to aid reassembly
- Create and maintain an asset register of all hardware in the workshop
- Document network configurations (IP addresses, MAC addresses, cable runs)
- Log software licences and installation keys
- Maintain a maintenance schedule for preventive tasks

EXAM TIP: Documentation is a professional requirement, not optional. In an exam context, know the contents of a service report: date, technician name, device description, fault reported, diagnosis, repair carried out, parts replaced, and sign-off confirmation.